IEEE INFOCOM 2022 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS) | 978-1-6654-0926-1/22/$31.00 ©2022 IEEE | DOI: 10.1109/INFOCOMWKSHPS54753.2022.9798090

Multi-frame Scheduling for Federated Learning over Energy-Efficient 6G Wireless Networks

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***Abstract*—It is envisioned that data-driven distributed learning approaches such as federated learning (FL) will be a key enabler for 6G wireless networks. However, the deployment of FL over wireless networks suffers from considerable energy consumption on communications and computation, which is challenging to meet stringent energy-efficiency goals of future sustainable 6G networks. In this paper, we investigate the energy consumption of transmitting scheduling decisions for FL deployed over a wireless network where mobile devices upload their local model to a coordinator (6G base station) for computing a global machine learning (ML) model iteratively. We consider that the coordinators have stringent energy efficiency goals. Therefore, to reduce the energy consumption due to the deployment of FL, we propose a novel *multi-frame* framework for FL that enables the coordinator to schedule wireless devices in one global round by only sending scheduling decisions at the beginning of each global round and setting the coordinator's transmission module to sleep mode to save power. In particular, we formulate a mixed-integer non-linear problem (MINLP) to minimize the average collection time of all device's local models by considering transmission errors. Then, we provide a novel method to solve the MINLP approximately and schedule wireless devices and allocate network resources. We demonstrate that our framework can save about 15 to 20 percent in some specific settings. Simulation results also show that our proposed algorithm outperforms traditional resource allocation methods and saves about 10% battery life per hundred global rounds in mobile device coordinators under certain scenarios.**

I. Introduction

Federated Learning (FL) was proposed in [1] to address the privacy concerns in distributed machine learning. In this approach, an FL system optimizes a global machine learning (ML) model by repeating the following process: i) each user device performs local computing with its data to minimize a global model. Then, they send their local model updates to the FL coordinator for aggregation; ii) the FL coordinator aggregates the received local model and produces an improved global model; iii) and the coordinator then sends the new updated global model to user devices, and user devices use this new global model as the reference. These steps create one global round, and global rounds are iterated until a desirable accuracy is reached.

FL can be deployed over wireless networks to exploit the increasing number of smart, powerful wireless devices such as mobile devices and Internet of Things (IoT) device. Due to its privacy preservation nature, FL is being treated as one of the vital solutions to achieve ubiquitous ML in 6G [2]. FL utilizes distributed data and computation power among users to solve complex problems using FL over wireless networks such as trajectory planning in Unmanned Aerial Vehicles (UAVs) [3], and edge content caching in base stations [4].

With all the promising benefits, FL comes with its own specific challenges when it is deployed over wireless networks. Firstly, user devices in wireless networks usually include unplugged devices and coordinators (6G base stations) with stringent energy-efficiency goals. Therefore, their power limitation should be considered in FL. Secondly, lossy nature of the wireless links can cause transmission errors, and therefore, affect the performance of FL. Thirdly, user devices' channel conditions differ from each other, and therefore, network resources should be allocated accordingly.

Most previous studies of FL over wireless networks utilize *frame-by-frame* scheduling where a global round consists of multiple frames each with multiple resource blocks (RBs) and after each frame, the coordinator sends scheduling decisions to allocate RBs to users for the next frame as shown in Figure 1a. It can take multiple frames for successful transmissions of local models to the coordinator before the coordinator can compute a global model. In the literature, it was always assumed that the coordinator has no constraint on energy consumption. However, energy-efficiency should be considered at coordinators for a sustainable 6G deployment.

To improve the energy-efficiency of coordinators for 6G wireless networks, we propose a novel *multi-frame* framework for FL that enables the coordinator to schedule user devices in multiple frames by only sending one scheduling decision at the beginning of each global round and setting the coordinator's transmission module to sleep mode to save power as shown in Figure 1b. We demonstrate that our framework enables the transmitter to save about 15 to 20 percent in energy consumption in transmitter sub-modules in low-power Wi-Fi modules (IEEE 802.11 b/g) by utilizing the built-in sleep mode. We understand Wi-Fi is not a cellular technology for 6G. However, considering the availability of radio specifications, in this paper, we use Wi-Fi to demonstrate the effectiveness of our proposed framework, which can be easily applied to 5G New Radio and beyond. Further, our multi-frame framework by considering transmission errors, schedules user devices in multiple frames such that it minimizes the collection time of

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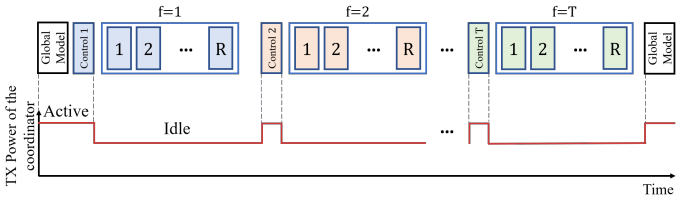
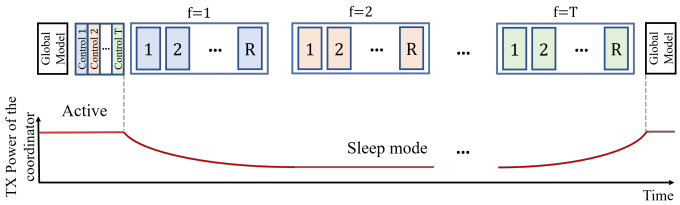


(a) Frame-by-frame scheduling



(b) Multi-frame scheduling

Fig. 1: Figure 1a) and 1b) show the frame structure and the corresponding transmission power of the coordinator. Due to high transition delay of the sleep mode, the coordinator cannot utilize sleep mode in the frame-by-frame scheduling.

all the user devices' local models for the FL process. The main contributions of the paper include:

- We introduce a novel multi-frame scheduling framework for FL over wireless networks with energy-limited coordinators. This framework can reduce the energy consumption of coordinators for 6G wireless networks without affecting the FL performance.
- By modeling the transmission errors, we formulate the collection time of user devices with a stochastic optimization problem with decision-dependent distribution and propose a novel algorithm to solve this mixed-integer non-linear problem (MINLP) approximately and schedule user devices and allocate network resources.
- We analytically prove that in a network with non-adjustable RB's bandwidth and no channel state information (CSI), the Round Robin scheduling policy minimizes the average collection time in the multi-frame framework.
- Finally, we provide experimental results to show that our proposed algorithm outperforms traditional scheduling algorithms in terms of average collection time.

The rest of this paper is organized as follows. In section II we discuss related works. The system model and problem formulation are described in section III. The joint user scheduling and resource allocation algorithm is introduced in section IV. In section V, we provide our experimental results and section VI concludes this paper.

II. Related Works

In this section, we describe recent works on challenges of deploying FL over wireless networks and real-world applications of energy-constrained FL coordinators.

A. FL Deployment over Wireless Networks

Recently, several works such as in [5]-[10] studied important problems related to the deployment of FL over wireless networks. More specifically, to address power limitation of user devices, authors in [5] proposed dynamically adjusting the CPU cycle frequency and transmission rate to minimize both power consumption and transmission time. However, this study does not consider the effect of transmission errors and energy consumption of requirement of coordinators. To address the effect of lossy nature of the wireless network on the performance of FL, authors in [6] utilized a noise-aware dynamic power transmission to maintain the same convergence rate of FL with perfect communications. However, a spectrum allocation method is not provided. To tackle the problem of various channel conditions, authors in [7] scheduled only a subset of user devices. However, aggregating only a subset of local models can slow down the convergence time and reduce model accuracy.

B. Energy-constrained FL Coordinators

Recently, [3] proposed FL for trajectory planning in a network of flying UAVs in which one of the UAVs is selected as a coordinator to improve network coverage and connectivity, and [8] utilized consensus protocols to select a mobile device from a network of mobile devices to act as a coordinator to develop serverless learning architectures for security improvements and solve the issue of single-point failures. For these unplugged coordinators, frame-by-frame scheduling consumes a lot of energy since the coordinator's transmission module has to be in the active state at all times compared to regular user devices which can go to sleep mode after their transmission. Since the time of each frame is very short and less than transition time between active and sleep mode, the coordinator's transmission module cannot go to sleep mode [9]. For instance, GS1011M low power Wi-Fi module from Gain Span has 15ms latency in wake time [10]. This reduction of power consumption during sleep modes is achieved by powering off power amplifiers [11].

III. System Model

We utilize the general model concept in [12] for our model. In particular, we consider a wireless network in which a set $\mathcal{K}$ of K users upload their local models in frames $f = \{1, 2, ..., T\}$ where T is the frame where all user's local models are collected successfully by the coordinator.

A. Transmission Model

For our uplink transmission model from user devices to the coordinator, we assume that orthogonal frequency-division multiple access (OFDMA) with adjustable frequency bandwidth is utilized because it is widely used in wireless devices. In our model, we assume that the total number of RBs that the coordinator can allocate to its users in each frame is R, and each user can occupy only one RB in each frame f. The total available bandwidth for each frame is b_{total} and the uplink rate of user $k \in \mathcal{K}$ that is transmitting its local FL model parameters to the coordinator at frame f is given by [13]:

$$c_k(\mathbf{r}_{k,f}, \mathbf{b}_f) = \sum_{n=1}^{R} r_{k,f}[n] b_f[n] \log_2(1 + \frac{P h_k}{I_n + b_f[n] N_0}), \quad (1)$$

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where $\mathbf{r}_{k,f} = [r_{k,f}[1], ..., r_{k,f}[n]]$ is an RB allocation vector with $r_{k,f}[n] \in \{0,1\}$; $r_{k,f}[n] = 1$ indicates that RB n is allocated to user k at the frame f, otherwise, we have $r_{k,f}[n] = 0$, and P is the transmit power of user k, which is assumed to be equal for all users, $\mathbf{b}_f = [b_f[1], ..., b_f[n]]$ is the bandwidth allocation vector for each RB in frame f with $b_f[n] \in [0, b_{total}]$; $b_f[n]$ indicates how much bandwidth RB n has, h_k is the channel gain between user k and the coordinator, N_0 is the noise power spectral density, and I_n is the the interference caused by others users.

Let G be the data size of the local FL model in bits. Therefore, the uplink transmission delay between user k and the coordinator at frame f is given by:

$$l_k(\mathbf{r}_{k,f}, \mathbf{b}_f) = \frac{G}{c_k(\mathbf{r}_{k,f}, \mathbf{b}_f)} \tag{2}$$

The time required for all selected users to upload their local models in frame f is give by:

$$t_f(\mathbf{a}_f, \mathbf{R}_f, \mathbf{b}_f) = \max_{k \in \mathcal{K}} a_f[k] l_k(\mathbf{R}_f, \mathbf{b}_f), \tag{3}$$

where $\mathbf{R}_f = [\mathbf{r}_{1,f}, ..., \mathbf{r}_{1,f}]$ and $\mathbf{a}_f = [a_f[1], ..., a_f[K]]$ is the user selection vector for the frame f and $a_f[k] \in \{0,1\}$. $a_f[k] = 0$ indicates that user k will not upload its local FL model to the coordinator at frame f and therefore $a_f[k] l_k(\mathbf{R}_f, \mathbf{b}_f) = 0$. $a_f[k] = 1$ means that user k will transmit its local model to the coordinator with transmission delay of $l_k(\mathbf{R}_f, \mathbf{b}_f)$. Equation (3) calculates the worst case transmission delay among all selected users in frame f.

B. Packet Error

Without losing generality, we assume that each local FL model is transmitted as a single packet in the uplink since if each local model require multiple sub-packet, in our model, this will translate to having multiple sub-user for each user that have those sub-packets. The error rate experienced by the transmission of user k's local FL model to the coordinator in frame f is as follows [14].

$$\mathcal{P}(E_{k,f} = 1) = e_{k,f}(\mathbf{r}_{k,f}, \mathbf{b}_f) = \sum_{n=1}^{R} r_{k,f}[n] e_{k,f}[n], \tag{4}$$

where $E_{k,f} \in \{0,1\}$ is the indicator variable of unsuccessful transmission; $E_{k,f} = 1$ indicates user k had unsuccessful transmission in frame f if it was scheduled and $e_{k,f}[n] = 1 - \exp\left(-\frac{v(I + b_f[n]N_0)}{Ph_k}\right)$ is the packet error rate of user k over RB n in frame f with v being a waterfall threshold [14].

We denote $Q_{k,f} = \{0,1\}$ to the indicator variable of the event of not receiving user k local model until end of frame f. $Q_{k,f} = 1$ indicates that user k have not successfully uploaded its local model to the coordinator until end of frame f. Otherwise, it is 0. $Q_{k,f}$ is calculates as follows.

$$Q_{k,f}(A, R, B) = \begin{cases} 1, & \text{w.p. } \bigcup_{i=1}^{f} \mathcal{P}(E_{k,i} = 1 | a_i[k] = 1) \\ 0, & \text{w.p. } 1 - \bigcup_{i=1}^{f} \mathcal{P}(E_{k,i} = 1 | a_i[k] = 1) \end{cases}, \tag{5}$$

where $A = [\mathbf{a}_1, ..., \mathbf{a}_f, ..., \mathbf{a}_T]$ is a user selection matrix for all the frames, $R = [\mathbf{R}_1, ..., \mathbf{R}_f, ..., \mathbf{R}_T]$ is an RB allocation matrix for all users at all the frames, and $B = [\mathbf{b}_1, ..., \mathbf{b}_f, ..., \mathbf{b}_T]$ is a bandwidth allocation matrix for all RBs in all the frames.

We denote $Z_f = \{0,1\}$ to the indicator variable of finishing receiving all the user devices local model successfully at the end of frame f. $Z_f = 1$ indicates that all the user devices have finished uploading their local models to the coordinator successfully in exactly frame f. Otherwise, it is 0. Z_k is calculated as follows.

$$Z_f(A, R, B) = \begin{cases} 1, \text{w.p.} & \bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,f-1} > 0) \\ & - \bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,f} > 0) \\ 0, \text{w.p.} & 1 - \bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,f-1} > 0) \\ & + \bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,f} > 0) \end{cases}, \tag{6}$$

where $\bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,0} > 0) = 1$. $Z_f = 1$ indicates that at least one of the users did not successfully upload their local model to the coordinator before frame f and all the users upload their local models to the coordinator at the end of frame f successfully. For abbreviation, in the rest of this paper, we denote $\mathcal{P}_f$ to $\bigcup_{k \in \mathcal{K}} \mathcal{P}(Q_{k,f} > 0)$.

C. Problem Formulation

We define a joint resource allocation and user selection scheme to minimize the time that all of the users require to upload their local updated model to the coordinator for global aggregation. The resource allocation and user selection problem is formulated as follows.

$$\min_{A,R,B} \quad \mathop{\mathbb{E}}_{T \sim Z_T = 1} \left[\sum_{f=1}^{T} t_f(\mathbf{a}_f, \mathbf{R}_f, \mathbf{b}_f) \right] \tag{7a}$$

$$\text{subject to} \quad a_f[k], r_{k,f}[n] \in \{0,1\} \quad \forall k \in \mathcal{K}, n = 1, ..., R, \tag{7b}$$

$$b_f[n] \in [0, b_{total}], \quad \forall n = 1, ..., R, \tag{7c}$$

$$\sum_{k \in \mathcal{K}} r_k[n] \leq 1, \quad \forall n = 1, ..., R, \tag{7d}$$

$$\sum_{n=1}^{R} b_f[n] = b_{total} \tag{7e}$$

$$\sum_{n=1}^{R} r_{k,f}[n] = a_f[k], \quad \forall k \in \mathcal{K} \tag{7f}$$

(7b) and (7d) imply that each user can only occupy one RB; (7c) and (7e) adjust the allocated bandwidth for each resource block and (7e) imply that all RBs must be allocated to the users that are associated with the coordinator. This optimization problem utilizes expectation value over T which is a random number representing the number of frames that user devices require to upload their local models to the coordinator successfully. T is a random value which comes from the distribution $Z_T = 1$.

This problems is a stochastic optimization problem with decision-dependent distribution. User scheduling and bandwidth allocation change the distribution of T, the number of frames that it is required to upload all the local models to the coordinator. Therefore, this problem is challenging to tackle.

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IV. Minimization of Transmission Time for Multi-frame Scheduling

To solve problem (7), we first consider the case where the bandwidth for all RBs are non adjustable and that all user devices have the same channel and power condition, such as a wireless network with RB fixed bandwidth and without CSI.

A. Fixed Bandwidth and Same User Conditions

In this part, we assume all users have the same channel and power condition, i.e., $h_k = \hat{h}$ and $I_n = \hat{I}$ for all $k \in \mathcal{K}$ and $n = 1, ..., R$. Also, we assume that all resource blocks have the same bandwidth in every frame, i.e., $b_f[n] = b_{total}/R = \hat{b}$ for all $n = 1, ..., R$. Since the channel conditions and bandwidth for each resource block is same, c_k is same across all devices, i.e., $c_k = \hat{c}$ for all $k \in \mathcal{K}$. Accordingly, the time that users require to upload their local model in each frame is same, i.e., $t_f = \hat{t}$ for each frame f. Also, $e_{k,f}$ will be the same for all user devices, frames, and resource blocks, i.e., $e_{k,f}[n] = \hat{e}$. In this scenario problem (7) will be as follows.

$$\min_{A} \quad \mathop{\mathbb{E}}_{T\sim Z_T=1}\left[\sum_{f=1}^{T}\hat{t}\right] \tag{8a}$$

$$\text{subject to} \quad a_f[k] \in \{0,1\} \quad \forall k \in \mathcal{K}, \tag{8b}$$

$$\sum_{k\in\mathcal{K}} a_f[k] = R, \tag{8c}$$

The main objective can be written as follows.

$$\begin{aligned}
&\mathop{\mathbb{E}}_{T\sim Z_T=1}\left[\sum_{f=1}^{T}\hat{t}\right] = \sum_{f=1}^{\infty}(f*\hat{t})\mathcal{P}(Z_f=1)\\
&= \sum_{f=1}^{\infty}(f*\hat{t})\left[\mathcal{P}_{f-1}-\mathcal{P}_f\right] = \hat{t}*\left[1+\sum_{f=1}^{\infty}\mathcal{P}_f\right]\\
&= \hat{t}*\left[1+\sum_{f=1}^{\infty}\left[\sum_{\forall k\in\mathcal{K}}\prod_{i=1}^{f}\hat{e}^{a_i[k]} - \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y}}\prod_{i=1}^{f}\hat{e}^{a_i[x]}\prod_{i=1}^{f}\hat{e}^{a_i[y]}\right.\right.\\
&\left.\left.+\sum_{\substack{\forall x,y,z\in\mathcal{K}\\ x\neq y\neq z}}\prod_{i=1}^{f}\hat{e}^{a_i[x]}\prod_{i=1}^{f}\hat{e}^{a_i[y]}\prod_{i=1}^{f}\hat{e}^{a_i[z]} - ...\right]\right.
\end{aligned}$$

where in the last equality, we used the fact that since transmission errors in different frames are independent of each other $\bigcup_{i=1}^{f}\mathcal{P}(E_{k,i}=1|a_i[k]=1) = \prod_{i=1}^{f}\mathcal{P}(E_{k,i}=1|a_i[k]=1)$ and $\prod_{i=1}^{f}\mathcal{P}(E_{k,i}=1|a_i[k]=1) = \prod_{i=1}^{f} e_{k,f}^{a_f[k]}$. Therefore, problem (8) is a binary integer convex problem which can be solved using a integer nonlinear programming (INLP) solver or exhaustive search. However, for this scenario it is possible to show that a round robin scheduling policy can achieve the minimum.

Theorem IV.1. *In a wireless network with fixed bandwidth RBs and without CSI, Round Robin scheduling policy minimizes the average transmission time for multi-frame scheduling at the coordinator.*

Proof. The average convergence time is $\hat{t}*\left[1+\sum_{f=1}^{\infty}\mathcal{P}_f\right]$ and we can show that $\mathcal{P}_f$ for any frame f gets minimized when using Round Robin scheduling policy. For this purpose, we denote $\sum_{i=1}^{f} a_i[k]$ to $\bar{a}_{k,f}$ and rewrite $\mathcal{P}_f$ as follows.

$$\begin{aligned}
&\sum_{\forall k\in\mathcal{K}}\hat{e}^{\bar{a}_{k,f}} - \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}} + \sum_{\substack{\forall x,y,z\in\mathcal{K}\\ x\neq y\neq z}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}}\hat{e}^{\bar{a}_{z,f}}\\
&- \sum_{\substack{\forall x,y,z,w\in\mathcal{K}\\ x\neq y\neq z\neq w}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}}\hat{e}^{\bar{a}_{z,f}}\hat{e}^{\bar{a}_{w,f}} + ...
\end{aligned} \tag{10}$$

If the resource allocation do not utilize Round Robin scheduling policy, it means that there exists two users $k^{'}$ and $k^{''}$ such that $\bar{a}_{k^{''},f} > \bar{a}_{k^{'},f} + 1$. We rewrite (10) as follows.

$$\begin{aligned}
&\hat{e}^{\bar{a}_{k^{'},f}}\overbrace{\left(1 - \sum_{\substack{\forall x\in\mathcal{K}\\ x\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}} + \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}} - ...\right)}^{\text{D}}\\
&- \hat{e}^{\bar{a}_{k^{'},f}}\hat{e}^{\bar{a}_{k^{''},f}}\left(1 - \sum_{\substack{\forall x\in\mathcal{K}\\ x\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}} + \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}} - ...\right)\\
&+ \hat{e}^{\bar{a}_{k^{''},f}}\left(1 - \sum_{\substack{\forall x\in\mathcal{K}\\ x\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}} + \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}} - ...\right)\\
&- \left(1 - \sum_{\substack{\forall x\in\mathcal{K}\\ x\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}} + \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y\neq k^{''}\neq k^{'}}}\hat{e}^{\bar{a}_{x,f}}\hat{e}^{\bar{a}_{y,f}} - ...\right) + 1\\
&= D*(\hat{e}^{\bar{a}_{k^{'},f}} - \hat{e}^{(\bar{a}_{k^{'},f}+\bar{a}_{k^{''},f})} + \hat{e}^{\bar{a}_{k^{''},f}} - 1) + 1\\
&= D*(\hat{e}^{\bar{a}_{k^{'},f}}(1+\hat{e}^{(\bar{a}_{k^{''},f}-\bar{a}_{k^{'},f})}) - \hat{e}^{(\bar{a}_{k^{'},f}+\bar{a}_{k^{''},f})} - 1) + 1
\end{aligned}$$

where $(\hat{e}^{\bar{a}_{k^{'},f}}(1+\hat{e}^{(\bar{a}_{k^{''},f}-\bar{a}_{k^{'},f})}) - \hat{e}^{(\bar{a}_{k^{'},f}+\bar{a}_{k^{''},f})} - 1)$ is a negative number and we need to maximize it. Therefore, $\bar{a}_{k^{''},f}$ should be the same or close to $\bar{a}_{k^{'},f}$, i.e., $\bar{a}_{k^{''},f} \leq \bar{a}_{k^{'},f}+1$ □

B. Dynamic and Fixed Bandwidth Resource Blocks

In this part, we will tackle the main resource allocation and user selection problem (7). Its main objective can be written as follows.

$$\begin{aligned}
&\mathop{\mathbb{E}}_{T\sim Z_T=1}\left[\sum_{f=i}^{T} t_f(\mathbf{a}_f, \mathbf{R}_f, \mathbf{b}_f)\right]\\
&= \sum_{f=1}^{\infty}\left(\sum_{i=1}^{f}\max_{k\in\mathcal{K}} a_i[k]\frac{G}{c_k(\mathbf{r}_{k,i}, \mathbf{b}_i)}\right)(\mathcal{P}_{f-1}-\mathcal{P}_f)\\
&= \sum_{f=1}^{\infty}\left(\max_{k\in\mathcal{K}} a_f[k]\frac{G}{\sum_{n=1}^{R} r_{i,f}[n]b_f[n]\log_2(1+\frac{Ph_k}{I_f+b_f[n]N_0})}\right)\mathcal{P}_{f-1}\\
&= \sum_{f=1}^{\infty}\left(\max_{k\in\mathcal{K}} a_f[k]\frac{G}{\sum_{n=1}^{R} r_{i,f}[n]b_f[n]\log_2(1+\frac{Ph_k}{I_f+b_f[n]N_0})}\right)\\
&*\left[\sum_{\forall k\in\mathcal{K}}\prod_{i=1}^{f-1} e_{k,f}^{a_i[k]} - \sum_{\substack{\forall x,y\in\mathcal{K}\\ x\neq y}}\prod_{i=1}^{f-1} e_{x,i}^{a_i[x]}\prod_{i=1}^{f-1} e_{y,i}^{a_i[y]} - ...\right]
\end{aligned}$$

Problem (12) in general is non-convex and it is a MINLP. We provide two methods to solve this problem. For the first method, we utilize MindtPy [15], an open-source software framework developed in Pyomo, which implements the Outer Approximation (OA) algorithm and the Extended Cutting Plane (ECP) algorithm.

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Algorithm 1 Greedy localized search algorithm

Parameters: S = number of sub frames, F = the total number of frames
for s = 0, 1, 2, ..., $S-1$ **do**
  $SubF = \{1 + s\frac{F}{S}, ..., (s+1)\frac{F}{S}\}$
  for each feasible solutions of $\mathbf{a}_{f \in SubF}$ **do**
    if RBs bandwiths are fixed **then**
      $\mathbf{b}_{f \in SubF} = b_{total}/R$
    else
      $\mathbf{b}_{f \in SubF} = \arg\min_{\mathbf{b}_f} \max_{k \in \mathcal{K}} a_f[k] \frac{G}{c_k(\mathbf{r}_{k,f}, \mathbf{b}_f)}$
    end if
    Calculate:
$$\sum_{f \in SubF'} t_f \mathcal{P}_{f-1} + t_{(s+1)\frac{F}{S}} \mathcal{P}_{(s+1)\frac{F}{S}},$$
    where $SubF' = \{SubF - \{(s+1)\frac{F}{S}\}\}$
    If the result is less than the previous least result, update $\mathbf{a}_{f \in SubF}$ and $\mathbf{b}_{f \in SubF}$.
  end for
  Output: $\mathbf{a}_{f \in SubF}$ and $\mathbf{b}_{f \in SubF}$
end for

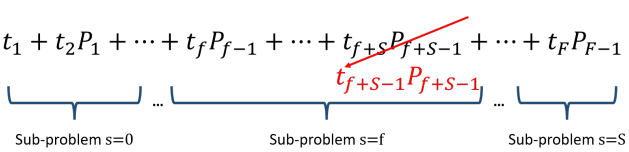

Fig. 2: Algorithm 1 splits equation (12) to independent sub-problems by approximating S-th variable of each sub-problem

Another method to solve problem (12) is to utilize exhaustive search for user allocation. However, this method is computationally heavy since this algorithm for F frames iterates $\binom{R}{K}^F$ times which is exponential and non-salable. For this purpose, we propose a greedy localized search algorithm that breaks down the F frames to S sub-frames and searches all the feasible solutions that minimize those sub-frames as a sub-problem for both fixed-bandwidth RB and dynamic RB. This algorithm is shown in Algorithm 1.

For fixed-bandwidth RBs, Algorithm 1 described in Figure 2 approximate FL learning time with fixed bandwidth by breaking down equation (12) to $\frac{F}{S}$ sub-problems by approximating S-th variable, i.e., $t_s(\mathbf{a}_s)\mathcal{P}_{s-1}$ with $t_{s-1}(\mathbf{a}_{s-1})\mathcal{P}_{s-1}$. By this approximation, it makes sure each sub-problem is independent of the next sub-problems and that these sub-problems can be solved in series. Algorithm 1's complexity is $S * \binom{R}{K}^{\frac{F}{S}}$. Increasing S will increase the accuracy of approximation with the cost of computational complexity where $S = F$ will solve the exact problem. On the other hand, $S = 1$ only considers one frame at a time, and it solves a greedy problem that only selects the user device with the best channel condition without considering the next frames and overall upload time.

For the dynamic bandwidth RBs, we extend the greedy localized search algorithm for fixed bandwidth. In this method, after breaking down the series to S sub-problems, in each allocation for each sub-problem, we solve the bandwidth allocation problem and find the best bandwidth for each RB that minimizes equation (12). With a fixed user schedule, the bandwidth allocation problem is non-convex, but it is possible to estimate it with a convex function by replacing $e_{k,f}[n]$ with its respective linear series expansion at $b_f[n] = 0$. With this approximation, given the user allocation, finding the best bandwidth for each user becomes a convex optimization problem that satisfies the KKT conditions. Therefore, this problem becomes equivalent to minimizing the transmission time of each frame, i.e., $\max_{k \in \mathcal{K}} a_i[k] \frac{G}{c_k(\mathbf{r}_{k,i}, \mathbf{b}_i)}$. This is because that $c_k(b_f[n])$ is more susceptible to changes of bandwidth because of its log function compared to $\mathcal{P}_f(\mathbf{b}_i)$. This process is described in Algorithm 1.

V. Simulation Results and Analysis

Parameter	Value	Parameter	Value
F	12	R	2
I	0.1 W	N_0	-174 dBm/Hz
B_{total}	5 MHz	P	1 W
v	1.2	G	10 Mb

TABLE I: System model parameters.

For our simulations, we consider a circular network area with a base station with total $b_{total} = 5$ MHz bandwidth, $R = 2$ resource blocks at the center, and K users distributed uniformly in the cell. We assume that all user devices have the same interference of $I = 0.2$ dBm, and that their channel condition does not change in the duration of each global round. We define $\gamma = \frac{K}{R}$ to represent the total number of users to the total number of available RBs in each frame. We assume each user's transmission power P_k is a uniform random number with a mean of 3 Watts and a range of change of σ to represent user devices heterogeneity. The other parameters used in simulations are listed in Table I.

Figure 3 shows the average transmission time of one global round of FL for network with and without adjustable RB bandwidth, $\sigma = 1$, and $F = 12$ using our proposed algorithms (Algorithm 1) and a naive Round Robin scheduling policy. From this figure, we see that as γ increases the transmission time increases. This is because the coordinator requires more frames to receive all of the users' packets. Also, this figure shows that our proposed algorithm performs better than the naive Round Robin scheduling algorithm. This is because that the proposed algorithm considers the transmission time of users and groups them for transmission in order to keep the time of each frame to the minimum and further, it schedules those frames in such a way that it would minimize the expectation value of the total number of frames. Figure 3 also shows that as the number of users increases, the proposed algorithm requires a larger parameter of S in order to consider allocation schemes where all the user can upload their packets. If S be a small number the algorithm only allocates resources to the user with better channel condition.

Figure 4 shows the average transmission time of one global round of FL for a network with $K = 4$ users, $R = 2$ RBs,

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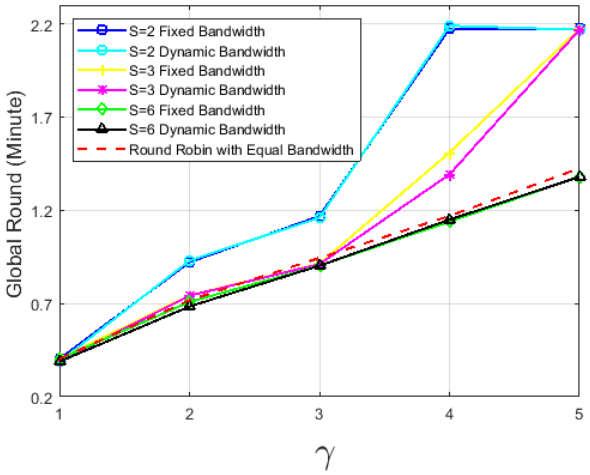

Fig. 3: Average transmission time of one global round in FL for a network with $\sigma = 1$ and varying γ.

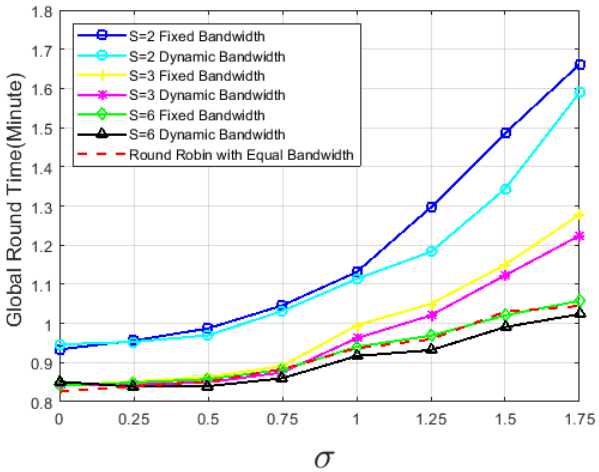

Fig. 4: Average transmission time of one global round in FL for a network with $K = 4$ and $R = 2$.

and overall frame of $F = 12$ and varying σ averaged over 30 samples. This figure shows as σ increases the expected transmission time increases since users with worse channel conditions create bottlenecks and increase the transmission time. However, dynamic RBs can compensate for bottlenecked users with lower transmission rates by allocating more bandwidth to these users to increase their transmission rate.

For calculating the energy saved using the multi-frame scheduling compared to frame-by-frame scheduling as shown in Figure 1, we consider a simple case where a mobile device with 3000 mAH battery is acting as the coordinator for $K = 50$ users at 1 Mbps transfer rate with perfect channels and $R = 1$ training an ML model of $G = 13$ Mbits. Using the previous studies and the available datasheets [16], [10], [17], we further assume that the network can save about 20 mA using the sleep method in multi-frame scheduling. Therefore, this mobile device is saving about 10% for each hundred global rounds of FL.

VI. Conclusion

In this paper, we proposed a novel multi-frame user selection and resource allocation framework that enables energy-limited coordinators to schedule user devices in multiple frames while minimizing the collection time of FL by considering transmission errors. We proposed an algorithm to both schedule users and allocate network resources and provided experimental results to show that our proposed algorithms outperforms traditional scheduling algorithms.

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